संप्रेषित-३.० २०२५
कल्पना की उड़ान, कला का संचार

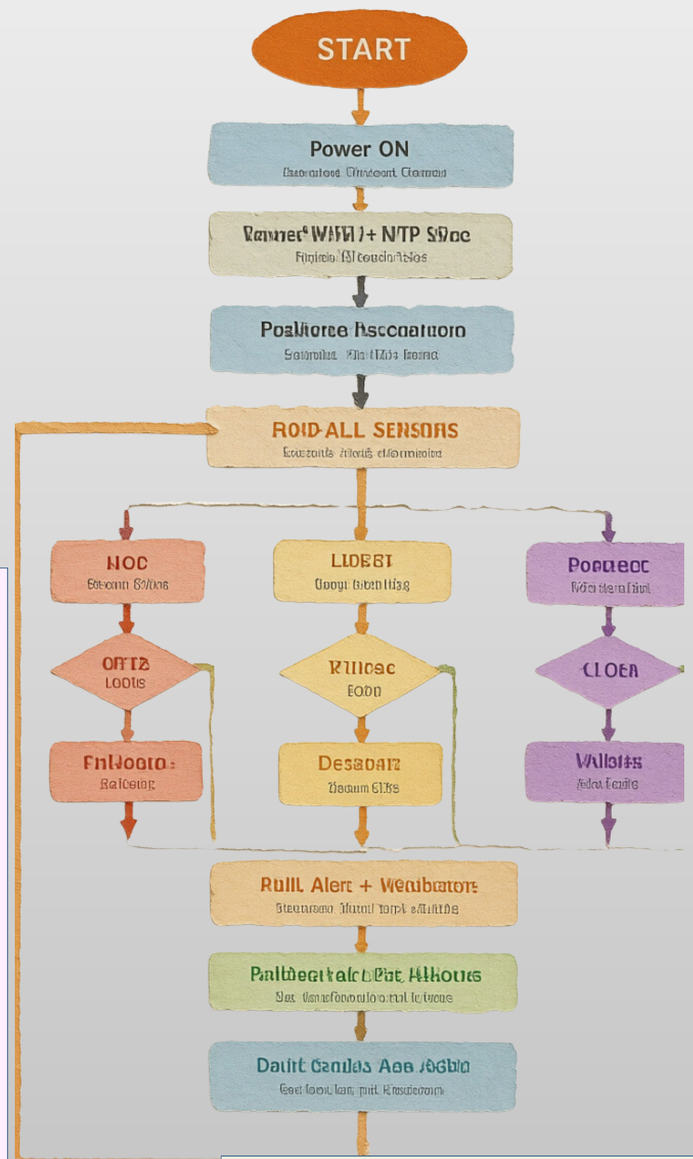
Branch:
ECS & EXTC
Group No. 6

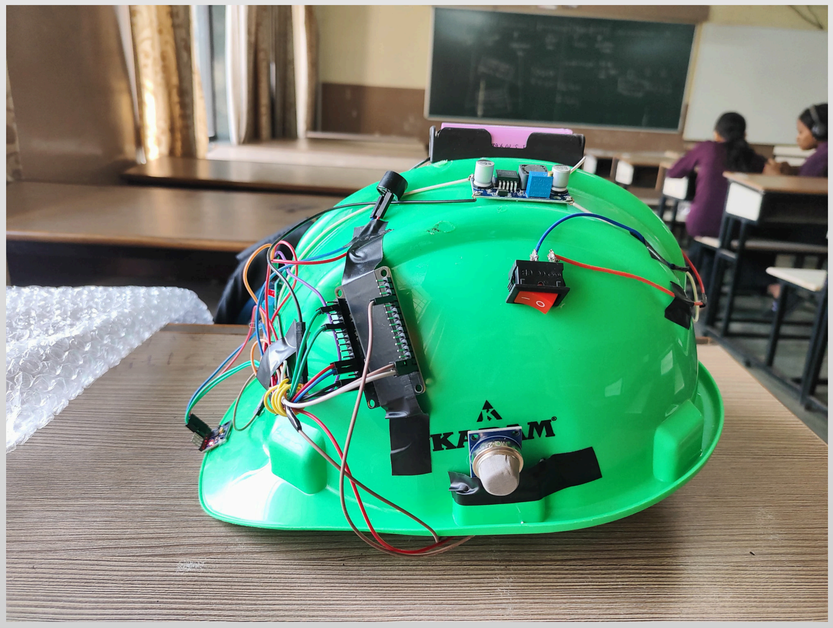
Shield-Sense

Smart Safety Construction Helmet

■HARDWARE USED

ESP32 (Main Controller)
MQ-2 Gas Sensor
DHT11 Temp & Humidity Sensor
MPU6050 (Fall Detection)
18650 Battery + Charging Module





●ESP32-based smart helmet
Real-time monitoring of:
~Gas levels
~Temperature & humidity
~Fall detection
~Firebase cloud integration
~Live web dashboard for supervisors

□ADVANTAGES

Improves worker safety
Real-time alerts
Easy monitoring through dashboard
Prevents accidents

■ APPLICATIONS

Construction Sites
Mining Industry
Industrial Plants
Fire & Rescue
Operation

● ALERT CONDITIONS

Gas Level > 400 → Warning
Temperature > 45°C
Humidity > 85%
Fall Detection via G-force

■LIMITATIONS

Requires internet connection
Battery dependent
Needs calibration of sensors

●IMPACT

Enhances worker safety
Enables real-time monitoring
Reduces accident &fatalities

●FUTURE IMPROVEMENTS

Improve battery efficiency
Build full web dashboard
Add offline functionality